MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

SILICON SAMPLING FOR INDIAN PUBLIC OPINION

How do LLM respondents use demographics? Prompting, fine-tuning and transfer

CAPSTONE PROJECT
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GROUP 55

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1. ABSTRACT / INTRODUCTION / MOTIVATION

Silicon sampling asks a language model to answer a survey as a respondent with a given demographic profile. It would be far cheaper than fielding a questionnaire. But is it faithful?

We test it in India on **Pew Religion in India** (N = 29,999) and **WVS-7 India** (N = 1,692), scoring every method on the same respondent-item rows as matched classifiers, and checking **direction**, **calibration** and **subgroup gaps**, not only accuracy.

2. LITERATURE REVIEW / BACKGROUND

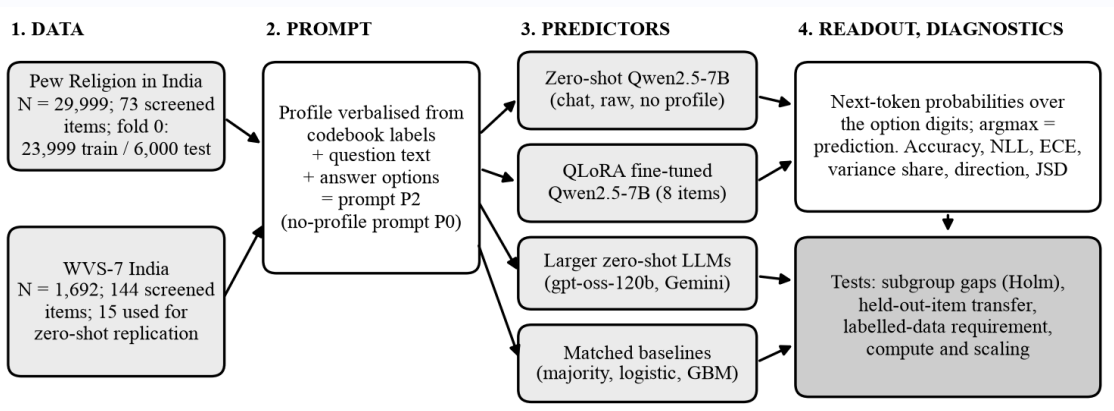
- LLMs reproduce US demographic-attitude correlations (Argyle 2023) but misrepresent many groups (Santurkar 2023).
- At the individual level they match or trail baselines and inflate gaps between groups (Chen 2026; Miranda 2025).
- Persona variables explain under 10% of variance (Hu 2024).
- Fine-tuning generalises to unseen subpopulations (Suh 2025).

Gap: India, what the profile itself contributes, and transfer to unseen questions.

3. OBJECTIVES

- **RQ1** How does a prompted LLM use the profile: how much, and in which direction?
- **RQ2** What does fine-tuning change versus zero-shot and matched classifiers?
- **RQ3** Does the adaptation transfer to unseen questions?
- **RQ4** Do a larger model or more labelled data change the picture?

4. METHODS



Same prompt and identical rows for every predictor: zero-shot Qwen2.5-7B (chat, raw, no profile), QLoRA fine-tuned Qwen2.5-7B, larger LLMs, and majority / logistic / GBM baselines.

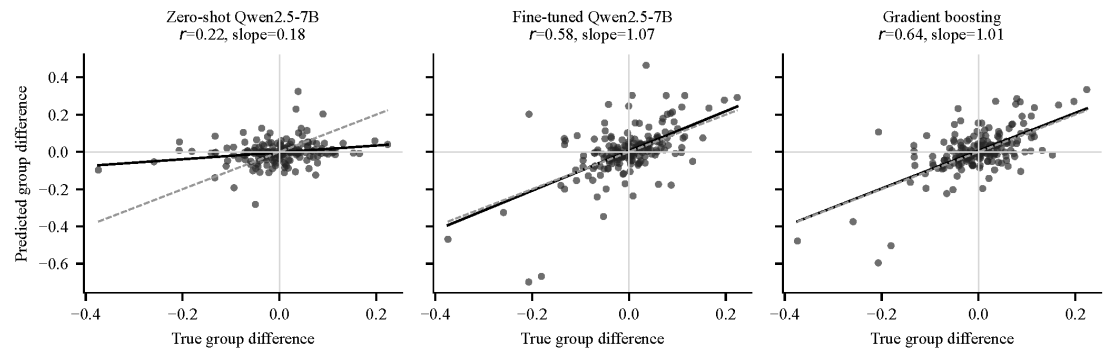
- **Data split**: 8 Pew items, five respondent-level folds; fold 0 = 23,999 train / 6,000 test respondents (46,209 test answers). Items 9 to 16 held out for transfer.
- **Readout**: the probability of each answer digit at the next token, so every method gives a full distribution, not just a label.
- **QLoRA**: 4-bit, rank 16, 10.1M trainable parameters, 184,853 examples, 6.6 GPU-hours.
- **Diagnostics**: accuracy, NLL, ECE, variance share, directional validity, JSD, Holm-corrected subgroup gaps.
- **Compute**: about 0.2 s per answer on a T4 GPU; classifiers are far cheaper.

5. RESULTS & DISCUSSION

17.5%
zero-shot accuracy vs 30.2% majority (200 respondents)

29% vs 4%
profile variance share: model vs real answers

38.0%
fine-tuned, 6,000 respondents (GBM 40.0%, ECE 0.009)



Predicted vs true group differences: zero-shot r = 0.22 (slope 0.18); fine-tuned r = 0.58; GBM r = 0.64.

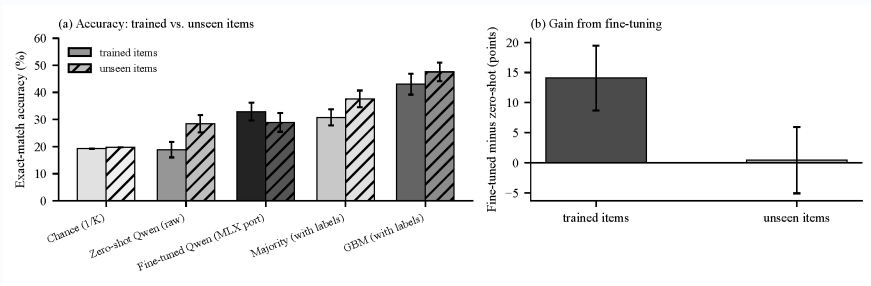
Pew fold 0 (46,209 answers)	Accuracy	NLL
Majority class	30.4%	1.59
Logistic regression	38.5%	1.47
Gradient boosting	40.0%	1.44
Fine-tuned Qwen2.5-7B	38.0%	1.48
Zero-shot Qwen2.5-7B (chat)	16.1%	4.26

- The profile is used **heavily but not usefully**: removing it raises zero-shot accuracy by 3.6 points.
- Fine-tuning recovers 79% of the gap to GBM, ending 2.1 points short.
- **No detectable gain on unseen items** (28.9% vs 28.5%, approximate port, 8 items).

Where it works, and where it does not

Slice of the test fold	Fine-tuned	GBM	Majority
Number of children (best item)	**56.4%**	57.5%	31.6%
Politicians in religion (weakest)	36.7%	39.4%	35.4%
Answers at ≥70% confidence (1,338)	**87.3%**	87.3%	23.2%

Confident answers are mostly unmarried respondents predicted to have no children, a rule a classifier also finds.



Held-out items: adapter vs base accuracy on 8 unseen Pew items (approximate port).

6. CONCLUSION & FUTURE SCOPE

- A prompted 7B model is **not** a faithful Indian respondent: it reacts to the profile strongly, in the wrong direction.
- QLoRA fine-tuning gives a calibrated simulator close to, but not above, matched classifiers.
- **Next**: all five folds and seeds, more held-out items, survey weights, and larger models on the same rows.

7. REFERENCES / ACKNOWLEDGEMENTS

Argyle et al., Political Analysis 31(3), 2023. • Santurkar et al., ICML 2023. • Chen, Zhu, Zheng, arXiv:2607.26348, 2026. • Miranda & Balbi, Entropy 27(9), 2025. • Hu & Collier, ACL 2024. • Suh et al., ACL 2025. • Dettmers et al., QLoRA, NeurIPS 2023.

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Code: github.com/DevilHunter695/silicon-sampling-capstone

TAKEAWAY A prompted model varies strongly with a profile but uses it weakly; fine-tuning closes most of the gap, and matched classifiers remain the benchmark.